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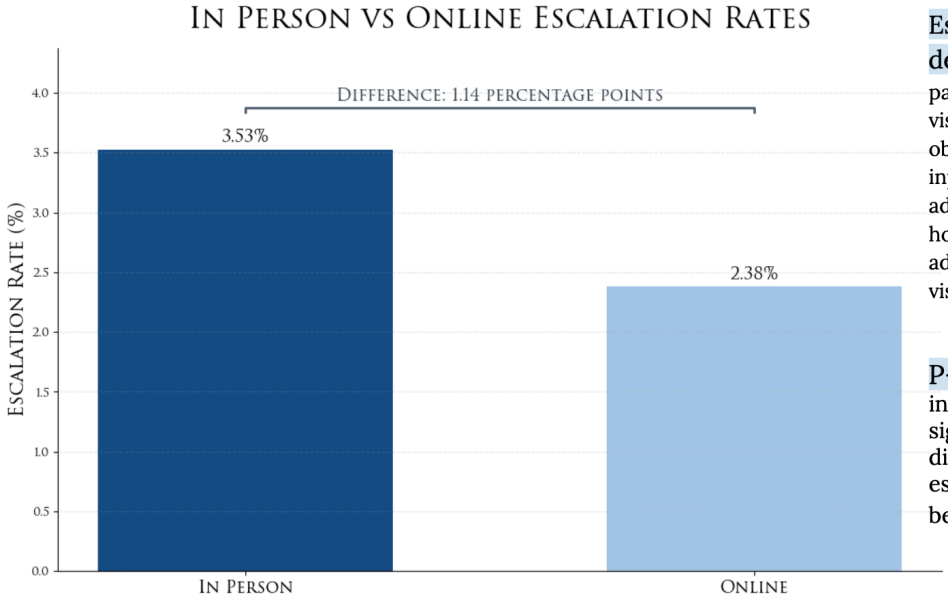
Patient Journeys Across In-person and Online Care

After obtaining the data, we decided to develop an understanding of a patient's journey across In-person and virtual care within the Stormont Vail Health centers using statistical analysis. We decided to go in the direction to determine how a patient's journey would be altered by if they did in-person visits or online virtual visits.

Our first question was if there was a significant difference in online patients versus in-person patients escalations. We defined escalation as a patient's future visits preceded by an online or in-person visit being observation, inpatient admission, hospital admission, or ED visit. We then sorted each patient's visits by date, identified the "next visit" after each baseline outpatient visit, checked if that next visit had any high-acuity flags, and created a next-visit escalation indicator which was used to determine the p-value. Once the calculations were completed we approximated that in-person visits had a 3.53% escalation rate, Online had a 2.38% rate, and there was an absolute difference of 1.14 favoring in-person. The p-value was very small and less than the significance rate of 0.05, so we concluded that the value indicated a significant difference in escalation rate between virtual and in-person visits, favoring in-person visits.

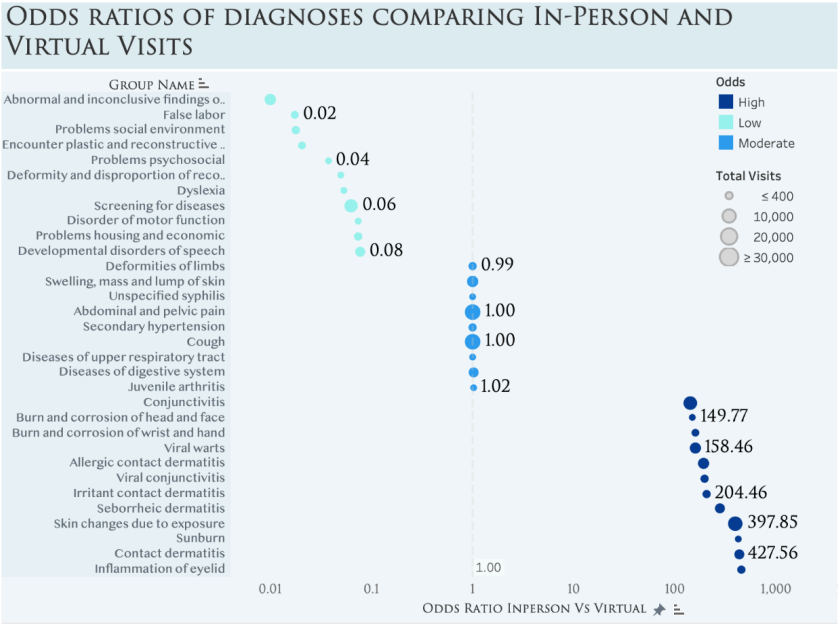
Subsequently we obtained odds-ratios for all diagnoses on the in-person vs online data. High odds ratio (>1) indicates a diagnosis is more likely to be an in-person visit. A low odds ratio (<1) indicates a diagnosis is more likely to be a virtual visit, and an odds ratio near 1 indicates no difference in if the diagnosis was more likely to be either virtual or an in-person visit. We ranked diagnoses by odds ratio and reviewed the top 20 highest values, lowest 20 values, and middle 20 values centered around 1. We reviewed trends and found that skin conditions made up 50% of the top 20 highest odds ratios (149.77 - 427.56). This indicates that skin conditions diagnoses are more likely to be in-person visits. Diagnoses associated with social environments such as problems with housing were typically likely to be virtual visits (0.019-.08 OR).

We concluded that future virtual visits could try to prioritize care for skin-conditions so that virtual visits could include better processes and technology for evaluating skin-conditions which would help the population that have a harder time participating in in-person visits. Additionally, improving virtual capabilities for assessing skin concerns could reduce barriers to timely care, increase patient satisfaction, and allow clinicians to triage cases more efficiently by identifying which patients truly need an in-person evaluation. Lastly, we would also advocate for assessing the root cause of why in-person visits had a higher preceding escalation rate versus virtual visits. This could potentially uncover ways for improvement for virtual care.



Escalation definition: patient's future visits being observation, inpatient admission, hospital admission, or ED visit

P-value < 0.05 indicates a significant difference in escalation rate between groups



Odds ratio:

High odds ratio (> 1) indicates a diagnosis is more likely to be in-person visit.

Low odds ratio (<1) indicates diagnosis more likely to be virtual visit

Trends:

Skin conditions made up 50% of the top 20 highest odds ratios (149 - 428) Indicated skin conditions diagnoses are more likely to be in-person visits

Diagnoses associated with social environment were typically likely to be virtual visits. (0.019-.08 OR)

